

TASK 1: File Identification

Objective

Identify the actual file type using its file signature.

Steps

- 1. Open **HxD**.
- 2. Click **File Open** and select the unknown file.
- 3. Observe the first 8–16 bytes.

Compare with Magic Numbers

File Type	Signature (Hex)
JPEG	FF D8 FF E0 / FF D8 FF E1
PNG	89 50 4E 47 0D 0A 1A 0A
PDF	25 50 44 46
ZIP	50 4B 03 04
EXE	4D 5A
GIF	47 49 46 38

Example

Header:

```
FF D8 FF E0
```

Result:

Actual File Type : JPEG
Extension : image.jpg
Status : Verified

TASK 2: Metadata Extraction

Objective

Extract EXIF metadata from a JPEG image.

Steps

1. Open JPEG file in HxD.
2. Press **Ctrl + F**.
3. Search for:

Exif

4. Locate EXIF block.
5. Search for keywords:

GPS
Canon
Nikon
Sony
DateTime
Model

Example Findings

Camera Model : Canon EOS 1500D
Date Taken : 2025:06:20 14:30:15
GPS Latitude : 28.6139
GPS Longitude : 77.2090

Evidence

Metadata	Value
Camera	Canon EOS 1500D
Date	2025-06-20
GPS	28.6139, 77.2090

TASK 3: Detecting Hidden Files

Objective

Identify files whose extension has been changed.

Example

File name:

holiday.jpg

Open in HxD and check header.

Header:

50 4B 03 04

Analysis:

50 4B 03 04 = ZIP Archive

Findings

File Name	Extension	Actual Type
holiday.jpg	JPG	ZIP Archive

Conclusion

Extension spoofing detected.
The file is actually a ZIP archive.

TASK 4: String Search

Objective

Search files for useful evidence.

Steps

1. Open file in HxD.
2. Press **Ctrl + F**.
3. Search as **Text-string**.

Search keywords:

admin
password
login
secret
gmail
http
https
username

Example Results

Username : siddharth
Password : Admin@123
URL : https://example.com
Email : user@gmail.com

Evidence Table

Offset	Evidence
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00000100	Username: siddharth
00000145	Password: Admin@123
00000200	URL: https://example.com
